Electric ($\mathbf{E}$; $\mathbf{D}$ in matter)

Magnetic ($\mathbf{B}$; $\mathbf{H}$ in matter)

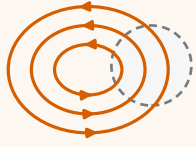
divergence: sources

$$\nabla \cdot \mathbf{D} = \rho_f$$

charges are flux sources

vacuum: $\nabla \cdot (\varepsilon_0 \mathbf{E}) = \rho$

Ch. 3 → Ch. 5



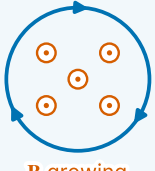
$$\nabla \cdot \mathbf{B} = 0$$

no flux source: in = out

B lines always close

Ch. 4

curl: circulation



$\mathbf{B}$ growing

$$\nabla \times \mathbf{E} = -\partial_t \mathbf{B}$$

changing B drives E loops

root of TMS induction

Ch. 4

$$\nabla \times \mathbf{H} = \mathbf{J}_f + \partial_t \mathbf{D}$$

currents drive B loops

vacuum: $\nabla \times (\mathbf{B}/\mu_0) = \mathbf{J} + \varepsilon_0 \partial_t \mathbf{E}$

Ch. 4 → Ch. 5

Consequences and companions: charge conservation (Ch. 3, 5) · interface conditions (Ch. 6) · material laws (Ch. 3, 5, 13)